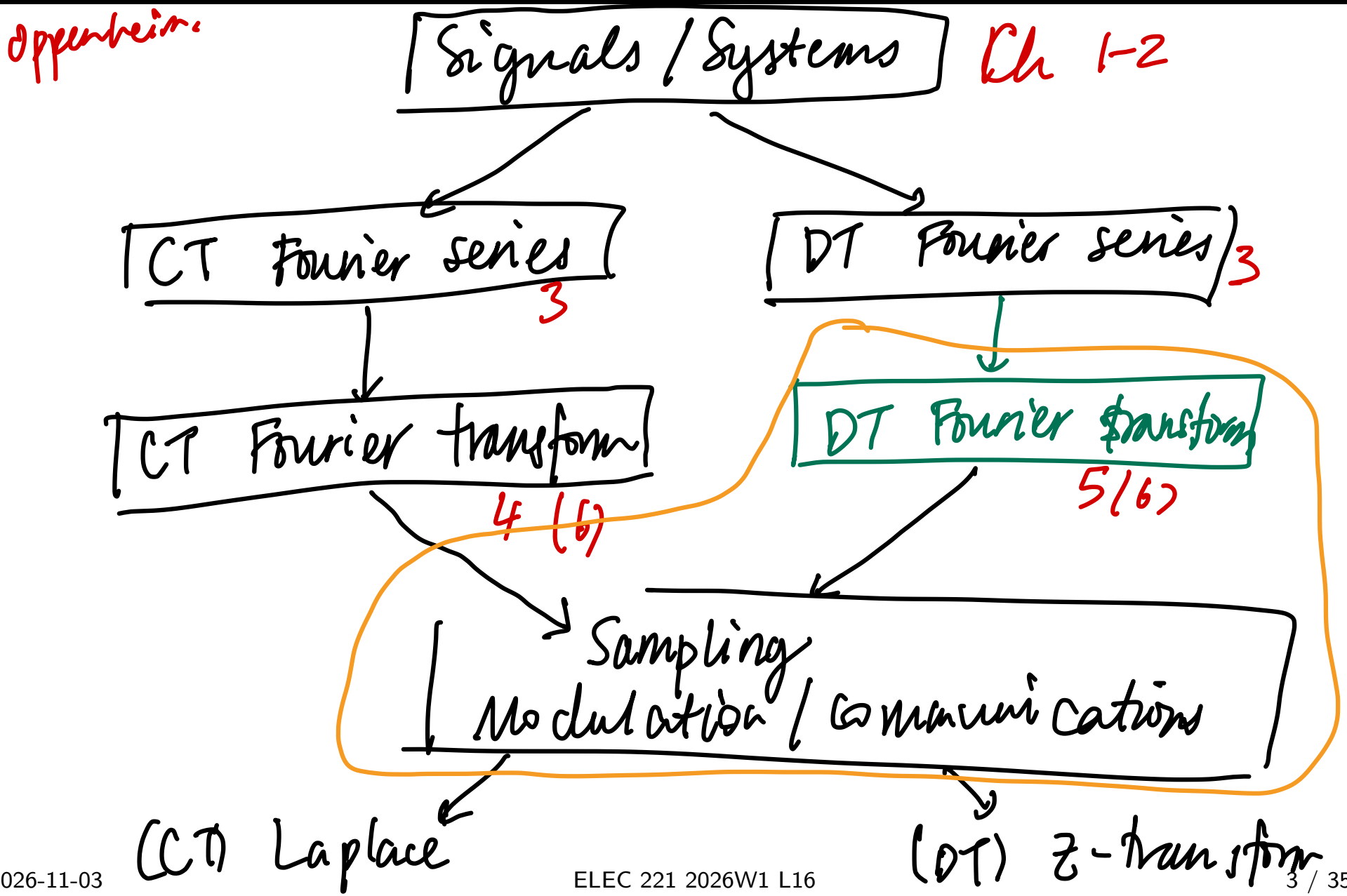
# ELEC 221 Lecture 16
# The DT Fourier transform

Tuesday 3 November 2026

# Announcements

- No quiz today!

# Where are we going?



# Today

Learning outcomes:

- Define and compute the DT Fourier transform for aperiodic signals
- Use the convolution property of the DTFT to analyze the behaviour of LTI systems
- Construct and analyze DT systems based on difference equations

# Recap: DT Fourier series

We can express a periodic DT signal (period $N$) as a discrete Fourier series.

DT synthesis equation:

$$x[n] = \sum_{k=\langle N \rangle} c_k e^{jk\frac{2\pi n}{N}}$$

DT analysis equation:

$$c_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\frac{2\pi n}{N}}$$

# The DT Fourier transform

The discrete-time Fourier transform (DTFT) is the generalization of the discrete Fourier series to **aperiodic** signals.

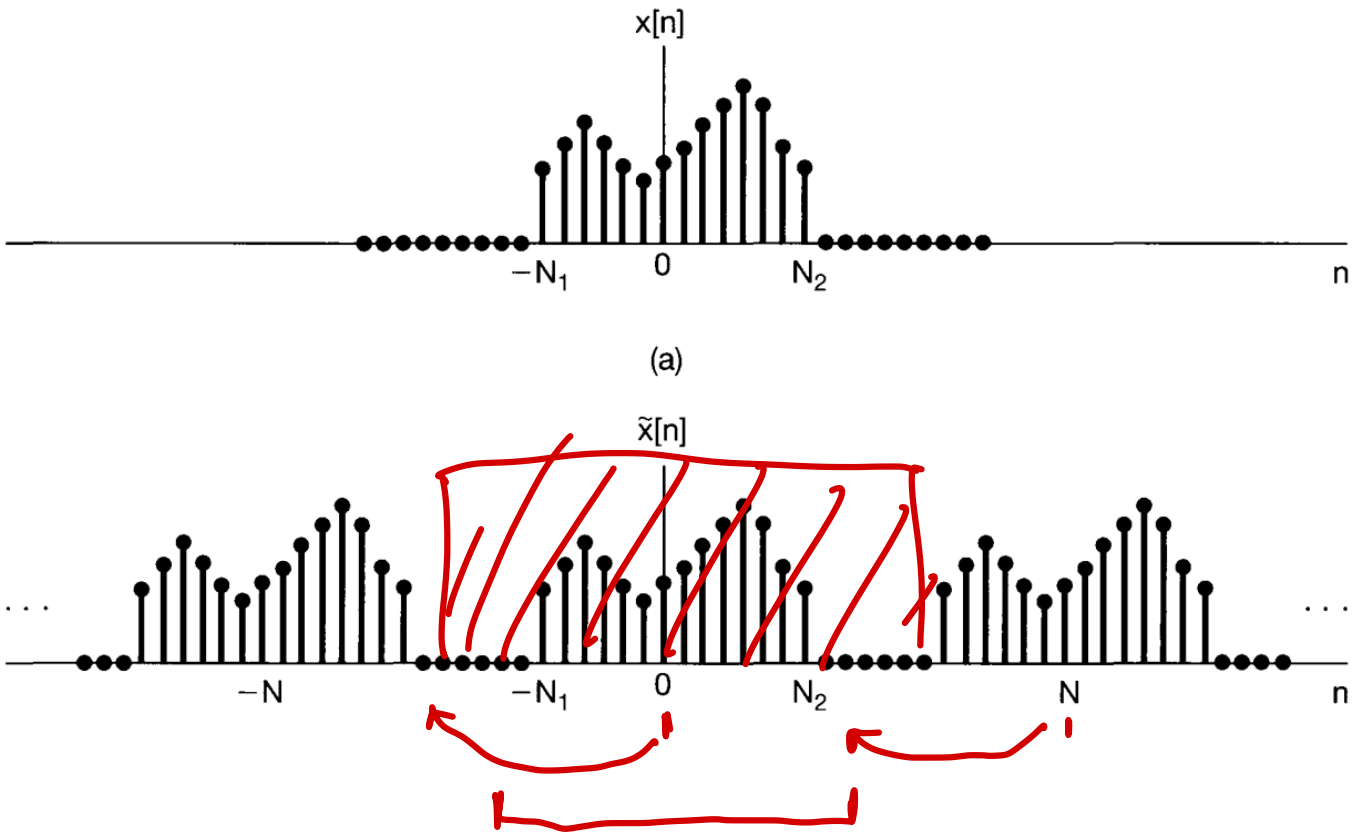



Image credit: Oppenheim chapter 5.1

# The DT Fourier transform

Suppose $\tilde{x}[n]$ is a periodic extension of $x[n]$.

$$\tilde{x}[n] = \sum_{k=\langle N \rangle} c_k e^{j k \frac{2\pi n}{N}} \qquad c_k = \frac{1}{N} \sum_{n=\langle N \rangle} \tilde{x}[n] e^{-j k \frac{2\pi n}{N}}$$

Set the bounds to consider where our signal actually is:

$$c_k = \frac{1}{N} \sum_{n=-N_1}^{N_2} \tilde{x}[n] e^{-j k \frac{2\pi n}{N}}$$

What happens if we increase the period?

# The DT Fourier transform

$$c_k = \frac{1}{N} \sum_{n=-N_1}^{N_2} \tilde{x}[n] e^{-jk\frac{2\pi n}{N}}$$

If $N \to \infty$, for any finite $n$, $\tilde{x}[n]$ looks just like $x[n]$:

$$c_k = \frac{1}{N} \sum_{n=-N_1}^{N_2} x[n] e^{-jk\frac{2\pi n}{N}}$$

Since $x[n] = 0$ outside this range, we can extend the bounds:

$$c_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk\frac{2\pi n}{N}}$$

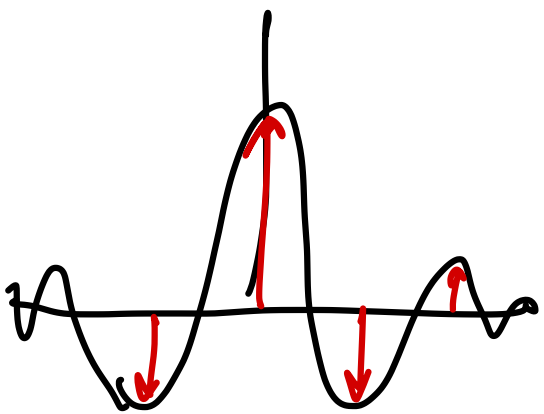
# The DT Fourier transform

We have

$$c_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk\frac{2\pi n}{N}}$$

Define

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \qquad \omega = \frac{2\pi}{N}$$



$$c_k = \frac{1}{N} X(e^{jk\omega})$$

# The DT Fourier transform

Substituting

$$c_k = \frac{1}{N} X(e^{jk\omega})$$

into the synthesis equation for $\tilde{x}[n]$ yields

$$\tilde{x}[n] = \sum_{k=\langle N \rangle} c_k e^{jk\omega n}$$

$$= \frac{1}{N} \sum_{k=\langle N \rangle} X(e^{jk\omega}) e^{jk\omega n}$$

$N = \frac{2\pi}{\omega}$

What happens as $N \to \infty$?

$$= \frac{1}{2\pi} \sum_{k=\langle N \rangle} X(e^{jk\omega}) e^{jk\omega n} \omega$$

$$\tilde{x}[n] = \frac{1}{2\pi} \int_0^{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \xrightarrow[N \to \infty]{} x[n]$$

Over what range should we integrate $\omega$?

# The DT Fourier transform

**DT Fourier transform pair**:

Inverse DTFT (synthesis equation)

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

DTFT (analysis equation)

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

notation (vs. $X(j\omega)$)

# Example: DTFT of a square pulse

Compute the DTFT of the DT signal

$$x[n] = \begin{cases} 1 & |n| \leq N_1 \\ 0 & |n| > N_1 \end{cases}$$

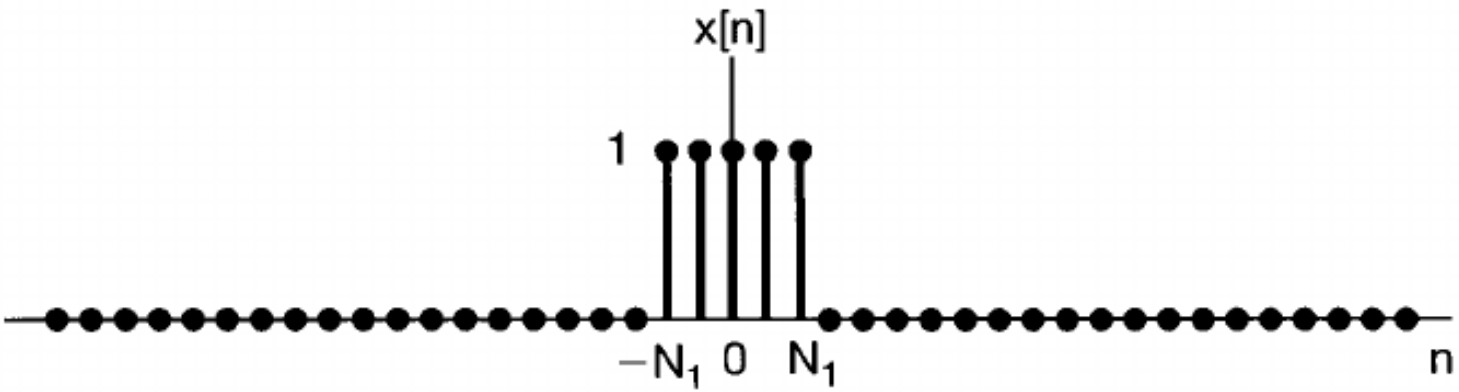



Image credit: Oppenheim chapter 5.1

# Recall: FT of a CT square pulse

$$x(t) = \begin{cases} 1 & |t| < T_1, \\ 0 & |t| > T_1 \end{cases}$$

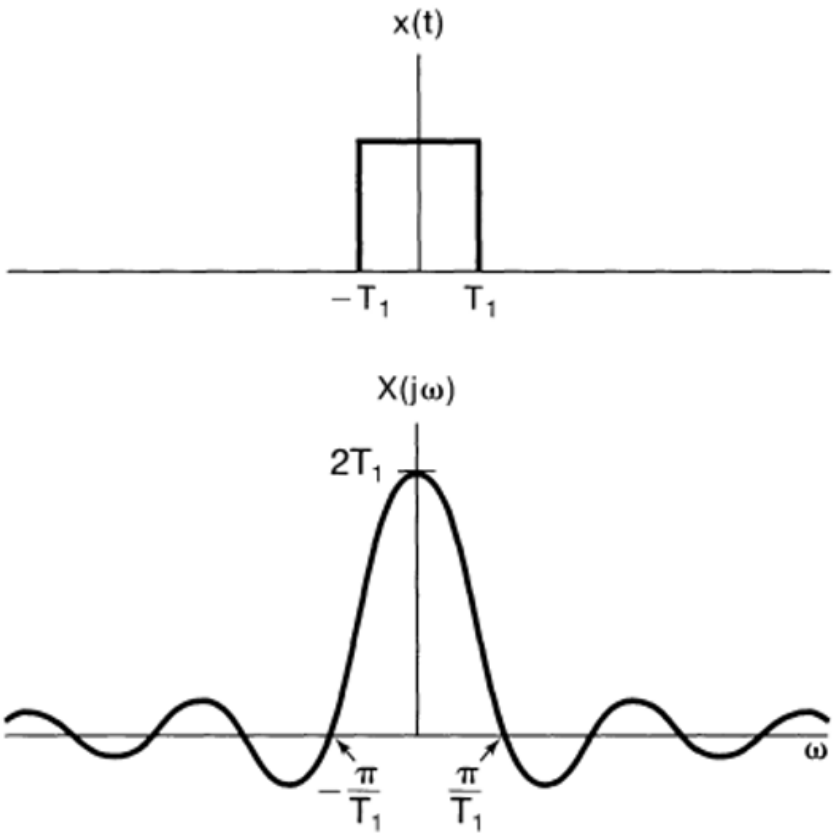



Image credit: Oppenheim chapter 4.1

# Example: DTFT of a square pulse

$$x[n] = \begin{cases} 1 & |n| \leq N_1 \\ 0 & |n| > N_1 \end{cases}$$

Compute the DTFT:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=-N_1}^{N_1} e^{-j\omega n}$$

How do we evaluate this sum?

# Example: DTFT of a square pulse

Change summation variable to $m = n + N_1$

$$X(e^{j\omega}) = \sum_{n=-N_1}^{N_1} e^{-j\omega n} = \sum_{m=0}^{2N_1} e^{-j\omega(m-N_1)} = e^{j\omega N_1} \sum_{m=0}^{2N_1} e^{-j\omega m}$$

Use our handy identity:

$$\sum_{k=0}^{N} z^k = \frac{1-z^{N+1}}{1-z} \Rightarrow X(e^{j\omega}) = e^{j\omega N_1}\left[\frac{1-e^{-j\omega(2N_1+1)}}{1-e^{-j\omega}}\right]$$

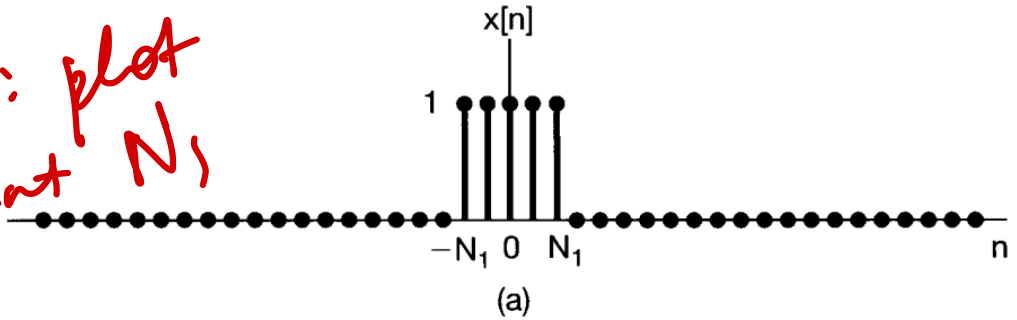
Do some reshuffling...

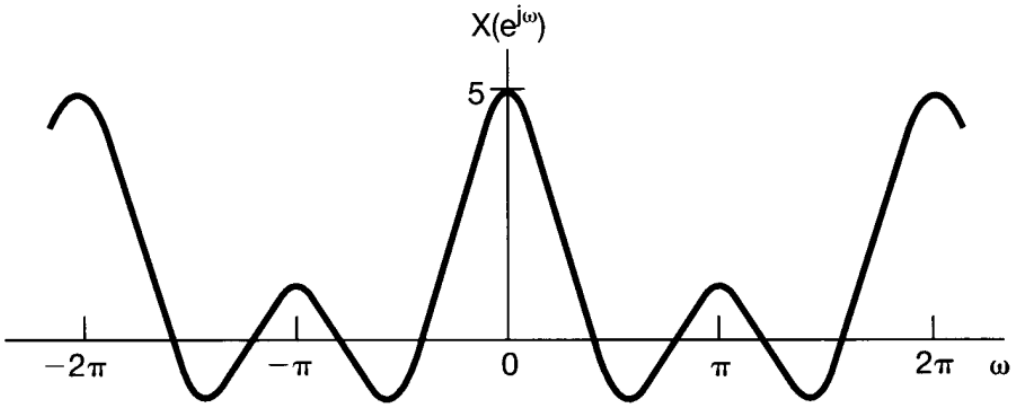
$$= e^{j\omega N_1} \frac{e^{-j\omega(N_1+1/2)}\left[e^{j\omega(N_1+1/2)} - e^{-j\omega(N_1+1/2)}\right]}{e^{-j\omega/2}\left[e^{j\omega/2} - e^{-j\omega/2}\right]}$$

$$X(e^{j\omega}) = \frac{\sin\left(\omega(N_1+1/2)\right)}{\sin(\omega/2)}$$

# Example: DTFT of a square pulse

Try yourself: plot for different $N_1$




Note this function is **continuous** and **periodic**!

Image credit: Oppenheim chapter 5.1

## Convergence criteria

In CT had Dirichlet criteria for both Fourier series and inverse Fourier transform. No conditions for DT Fourier series:

$$x[n] = \sum_{k=\langle N \rangle} c_k e^{jk\omega n} \qquad c_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega n}$$

What about the DT Fourier transform?

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \qquad X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

One of the following must hold:

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty \qquad \sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty$$

* Look into counterex. or justification.

## Convolution

DTFT defines the relationship between impulse response and frequency response:

$$h[n] \overset{\mathcal{F}}{\longleftrightarrow} H(e^{j\omega})$$

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n}$$

$$h[n] = \frac{1}{2\pi} \int_{2\pi} H(e^{j\omega}) e^{j\omega n} d\omega$$

Convolution works the same way as in CT:

$$y[n] = h[n] * x[n]$$

$$Y(e^{j\omega}) = H(e^{j\omega}) X(e^{j\omega})$$

# Example: filters

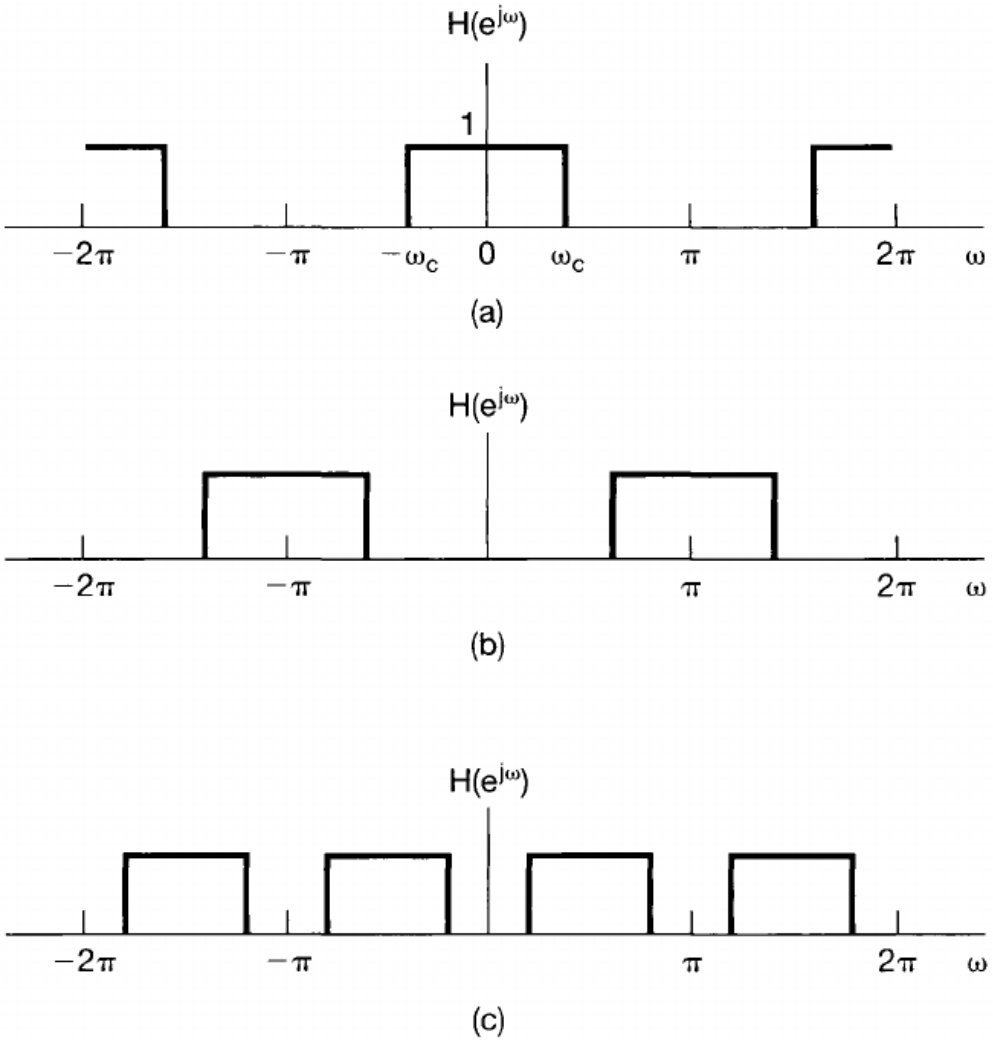



Image: Oppenheim 3.28

## Example: filters

Determine the impulse response of an ideal DT low-pass filter,

$$H(e^{j\omega}) = \begin{cases} 1, & 0 \leq |\omega| < \omega_c \\ 0, & \omega_c \leq |\omega| < \pi \end{cases}$$

$$h[n] = \frac{\sin(\omega_c n)}{\pi n}$$

## Example: filters

For an ideal DT high-pass filter,

# Properties of the DT Fourier transform

**TABLE 5.1** PROPERTIES OF THE DISCRETE-TIME FOURIER TRANSFORM

| Section | Property | Aperiodic Signal | Fourier Transform |
|---|---|---|---|
| | | $x[n]$ | $X(e^{j\omega})$ periodic with |
| | | $y[n]$ | $Y(e^{j\omega})$ period $2\pi$ |
| 5.3.2 | Linearity | $ax[n] + by[n]$ | $aX(e^{j\omega}) + bY(e^{j\omega})$ |
| 5.3.3 | Time Shifting | $x[n - n_0]$ | $e^{-j\omega n_0}X(e^{j\omega})$ |
| 5.3.3 | Frequency Shifting | $e^{j\omega_0 n}x[n]$ | $X(e^{j(\omega-\omega_0)})$ |
| 5.3.4 | Conjugation | $x^*[n]$ | $X^*(e^{-j\omega})$ |
| 5.3.6 | Time Reversal | $x[-n]$ | $X(e^{-j\omega})$ |
| 5.3.7 | Time Expansion | $x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n = \text{multiple of } k \\ 0, & \text{if } n \neq \text{multiple of } k \end{cases}$ | $X(e^{jk\omega})$ |
| 5.4 | Convolution | $x[n] * y[n]$ | $X(e^{j\omega})Y(e^{j\omega})$ |
| 5.5 | Multiplication | $x[n]y[n]$ | $\frac{1}{2\pi}\int_{2\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$ |
| 5.3.5 | Differencing in Time | $x[n] - x[n-1]$ | $(1 - e^{-j\omega})X(e^{j\omega})$ |
| 5.3.5 | Accumulation | $\sum_{k=-\infty}^{n} x[k]$ | $\frac{1}{1-e^{-j\omega}}X(e^{j\omega})$ $+\pi X(e^{j0})\sum_{k=-\infty}^{+\infty}\delta(\omega - 2\pi k)$ |
| 5.3.8 | Differentiation in Frequency | $nx[n]$ | $j\frac{dX(e^{j\omega})}{d\omega}$ |
| 5.3.4 | Conjugate Symmetry for Real Signals | $x[n]$ real | $\begin{cases} X(e^{j\omega}) = X^*(e^{-j\omega}) \\ \mathcal{R}e\{X(e^{j\omega})\} = \mathcal{R}e\{X(e^{-j\omega})\} \\ \mathcal{I}m\{X(e^{j\omega})\} = -\mathcal{I}m\{X(e^{-j\omega})\} \\ \lvert X(e^{j\omega})\rvert = \lvert X(e^{-j\omega})\rvert \\ \measuredangle X(e^{j\omega}) = -\measuredangle X(e^{-j\omega}) \end{cases}$ |
| 5.3.4 | Symmetry for Real, Even Signals | $x[n]$ real an even | $X(e^{j\omega})$ real and even |
| 5.3.4 | Symmetry for Real, Odd Signals | $x[n]$ real and odd | $X(e^{j\omega})$ purely imaginary and odd |
| 5.3.4 | Even-odd Decomposition of Real Signals | $x_e[n] = \mathcal{E}v\{x[n]\}$ $[x[n]$ real$]$<br>$x_o[n] = \mathcal{O}d\{x[n]\}$ $[x[n]$ real$]$ | $\mathcal{R}e\{X(e^{j\omega})\}$<br>$j\mathcal{I}m\{X(e^{j\omega})\}$ |
| 5.3.9 | Parseval's Relation for Aperiodic Signals | $\sum_{n=-\infty}^{+\infty}\lvert x[n]\rvert^2 = \frac{1}{2\pi}\int_{2\pi}\lvert X(e^{j\omega})\rvert^2 d\omega$ | |

Image credit: Oppenheim Table 5.1

# Properties of the DT Fourier transform

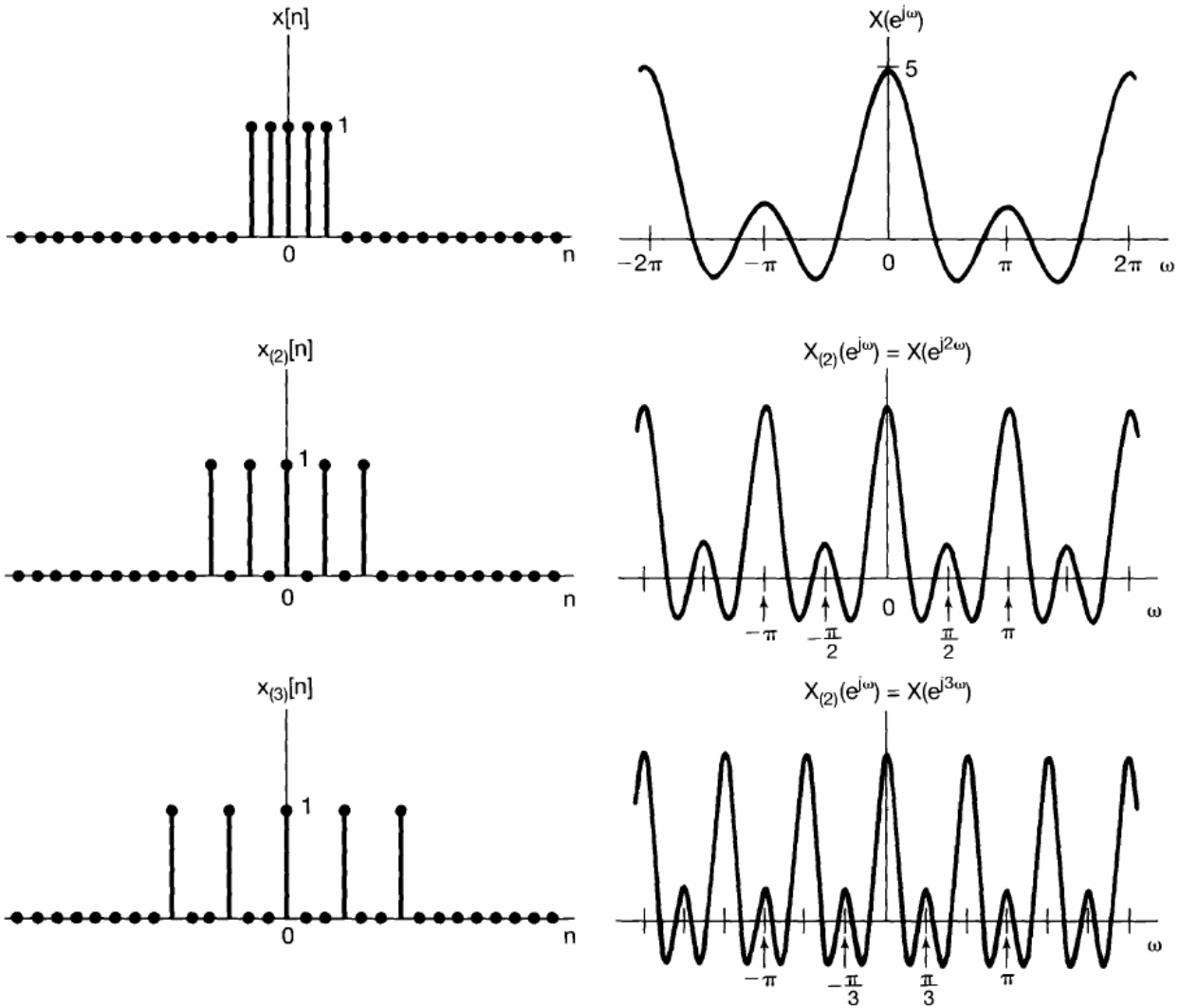
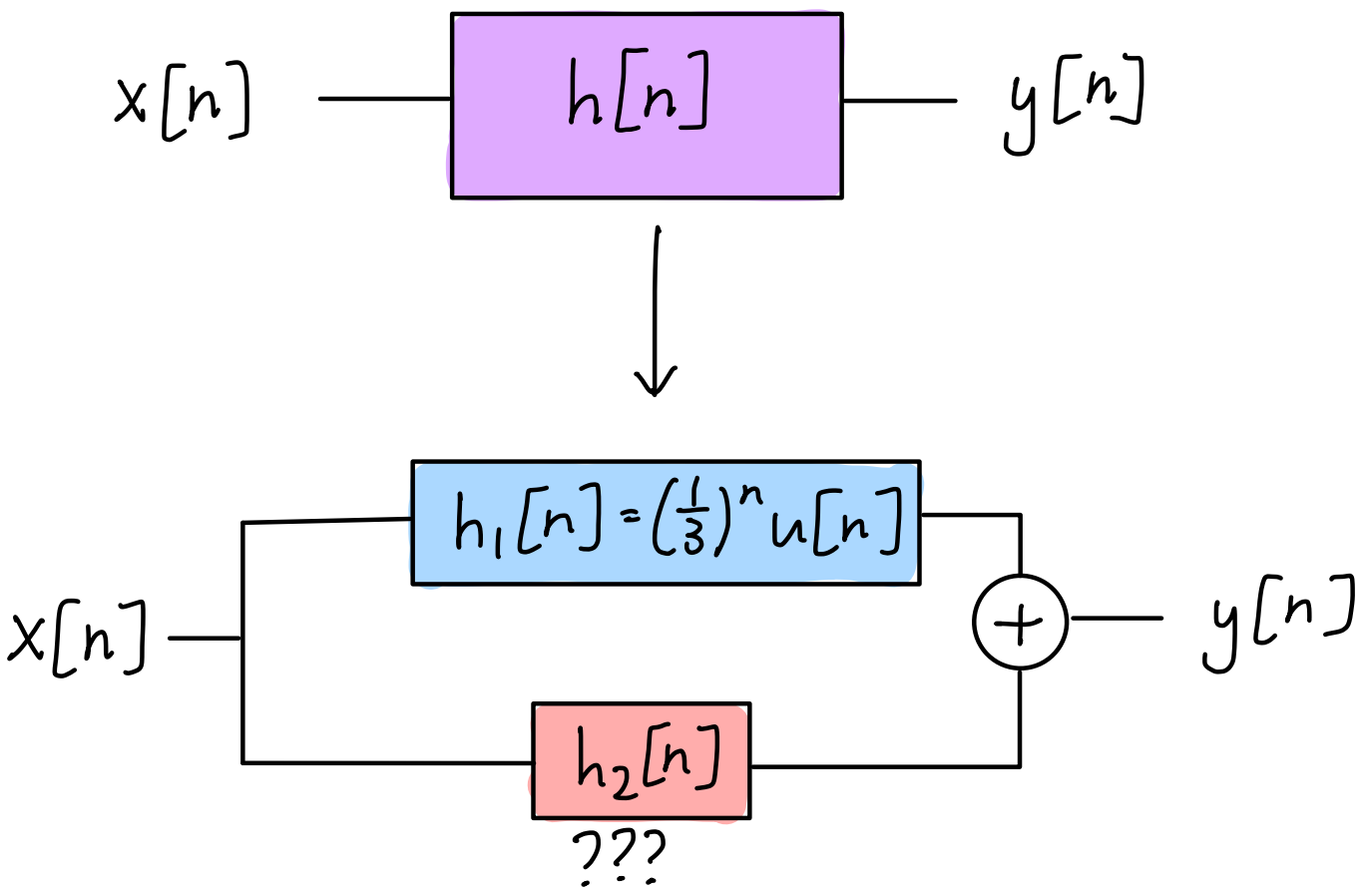



Image credit: Oppenheim Figure 5.14

# Example

$$H(e^{j\omega}) = \frac{-12 + 5e^{-j\omega}}{12 - 7e^{-j\omega} + e^{-2j\omega}}$$



# Example

What is the DTFT of

$$x[n] = a^n u[n], \quad |a| < 1$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} a^n e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} (a e^{-j\omega})^n$$

$$= \frac{1}{1 - a e^{-j\omega}}$$

$x(t) = e^{-at} u(t)$ $\leftrightarrow$ $\frac{1}{a + j\omega}$ $\quad Re(a) > 0$

# Example

$$h[n] = h_1[n] + h_2[n] = \left(\frac{1}{3}\right)^n u[n] + h_2[n]$$

$$\rightarrow \quad H_1(e^{j\omega}) = \frac{1}{1 - \frac{1}{3}e^{-j\omega}}$$

$$\Rightarrow H(e^{j\omega}) = \frac{-12 + 5e^{-j\omega}}{12 - 7e^{-j\omega} + e^{-2j\omega}}$$

$$\vdots$$

find $H_2(e^{j\omega})$ s.t. $H(e^{j\omega}) = H_1(e^{j\omega}) + H_2(e^{j\omega})$

we'll finish up the example next time.

## For next time

Content:

- The sampling theorem
- Basics of interpolation
- The Nyquist rate and aliasing

Action items:

1. Enjoy not having to study for midterm

Recommended reading:

- From this class: Oppenheim 5.2-5.9
- Suggested problems: 5.4b, 5.6, 5.8, 5.19, 5.22bcdfgh, 5.25, 5.29, 5.31, 5.33-5.36
- For next class: Oppenheim 7.1-7.3